

Hardware Components List & Procurement Checklist

Project: Real-Time WESAD ECG Telemetry & Autonomic Stress Detection System (STM32G474 + AD8232)

1. PRIMARY PROCESSING & COMPUTING HARDWARE

☐	Component	Recommended Part / Spec	Description & Purpose
☐	Microcontroller Board	ST NUCLEO-G474RE	ARM Cortex-M4 @ 170 MHz with FPU. Runs on-chip Biquad DSP filter and CRC-16 telemetry packing. Includes on-board ST-LINK V3 programmer/debugger and Virtual COM Port.
☐	Host Computer	Windows 10 / 11 Laptop	Runs STM32CubeIDE for firmware flashing, the Python serial ingestion engine, and the interactive Streamlit telemetry dashboard.
☐	USB Interface Cable	Micro-USB to USB-A (or USB-C)	Connects Nucleo board to laptop. Must be a full data-sync cable (with D+ and D- data lines), not a charge-only phone cord.

2. BIOSIGNAL ANALOG FRONT-END (AFE)

☐	Component	Recommended Part / Spec	Description & Purpose
☐	ECG Sensor Breakout	AD8232 Heart Rate Monitor	Analog front-end instrumentation amplifier with integrated high-pass filter and Right-Leg Drive (RLD) to extract clean ~1 mV cardiac microvolt signals from skin.
☐	Biomedical Sensor Cable	3.5 mm Jack to 3 Snap Leads	Shielded cable connecting the AD8232 3.5 mm input socket to 3 snap electrodes: Red (RA), Yellow/Black (LA), and Green (RL reference).
☐	Disposable ECG Electrodes	Ag/AgCl Pre-Gelled Foam (10-30 pcs)	Standard medical snap electrodes (e.g., 3M Red Dot or generic standard ECG pads) for low-impedance electrical skin contact.

3. WIRING, HEADERS & PROTOTYPING

☐	Component	Recommended Part / Spec	Description & Purpose
☐	DuPont Jumper Wires	Female-to-Male (F-M), ~20 cm	At least 6 wires: Connects AD8232 pins (3.3V, GND, OUTPUT, LO-, LO+,

☐	Component	Recommended Part / Spec	Description & Purpose
			SDN) directly to STM32 Nucleo headers.
☐	Male-to-Male Jumpers	M-M Jumper Pack (~10 pcs)	Useful for breadboard routing or adding external test points.
☐	Male Header Pins	0.1" (2.54 mm) Pitch, 1x6 Strip	To solder onto the AD8232 board if it arrives unpopulated without header pins.
☐	Solderless Breadboard	Half-size (400 tie-points)	Provides physical stability for the sensor module during bench testing.

Critical Clinical Safety Rule: Always run the host laptop on **battery power (unplugged from AC wall outlet)** while electrodes are attached to a human subject. This prevents 50 Hz powerline ground loops and guarantees complete galvanic isolation.

Skin Preparation Tip: Clean electrode placement sites with standard **70% Isopropyl Alcohol prep swabs** prior to applying electrodes. Removing dead skin cells and oils reduces contact impedance by over 80%, virtually eliminating baseline wander.